Satellite Land Cover Change Detection

Analysis Report · EuroSAT / ResNet50 / Sentinel-2

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OVERVIEW

A ResNet50 model fine-tuned on the EuroSAT dataset classifies each 64×64 pixel patch of a Sentinel-2 scene into one of 10 land-cover types. By comparing classifications from two time periods, the pipeline flags patches where forest or vegetation was replaced by agriculture, pasture, or urban land use acting as proxies for deforestation and urban encroachment.

Three geographically distinct study areas were selected to test the model under different biomes, land-use pressures, and urban contexts: the Atlantic Forest and Amazon frontiers of Brazil, and the peri-urban fringe of Karachi, Pakistan.

RESULTS SUMMARY

Study Area	Period	Flagged Patches
Paraná, Brazil	2018 → 2024	28 (Forest to AnnualCrop / Pasture)
Mato Grosso, Brazil	2018 → 2026	10 (Forest to AnnualCrop / Pasture)
Karachi, Pakistan	2018 → 2024	6 (HerbVeg to Residential / Industrial)

MODEL PERFORMANCE

ResNet50 fine-tuned on EuroSAT — test set (2,700 images)

Training used a two-phase approach: first only the final classification head was trained (frozen backbone), then the full network was unfrozen and fine-tuned at a 10× lower learning rate. Best checkpoint by validation accuracy was saved after each epoch; early stopping with patience=3 prevented overfitting.

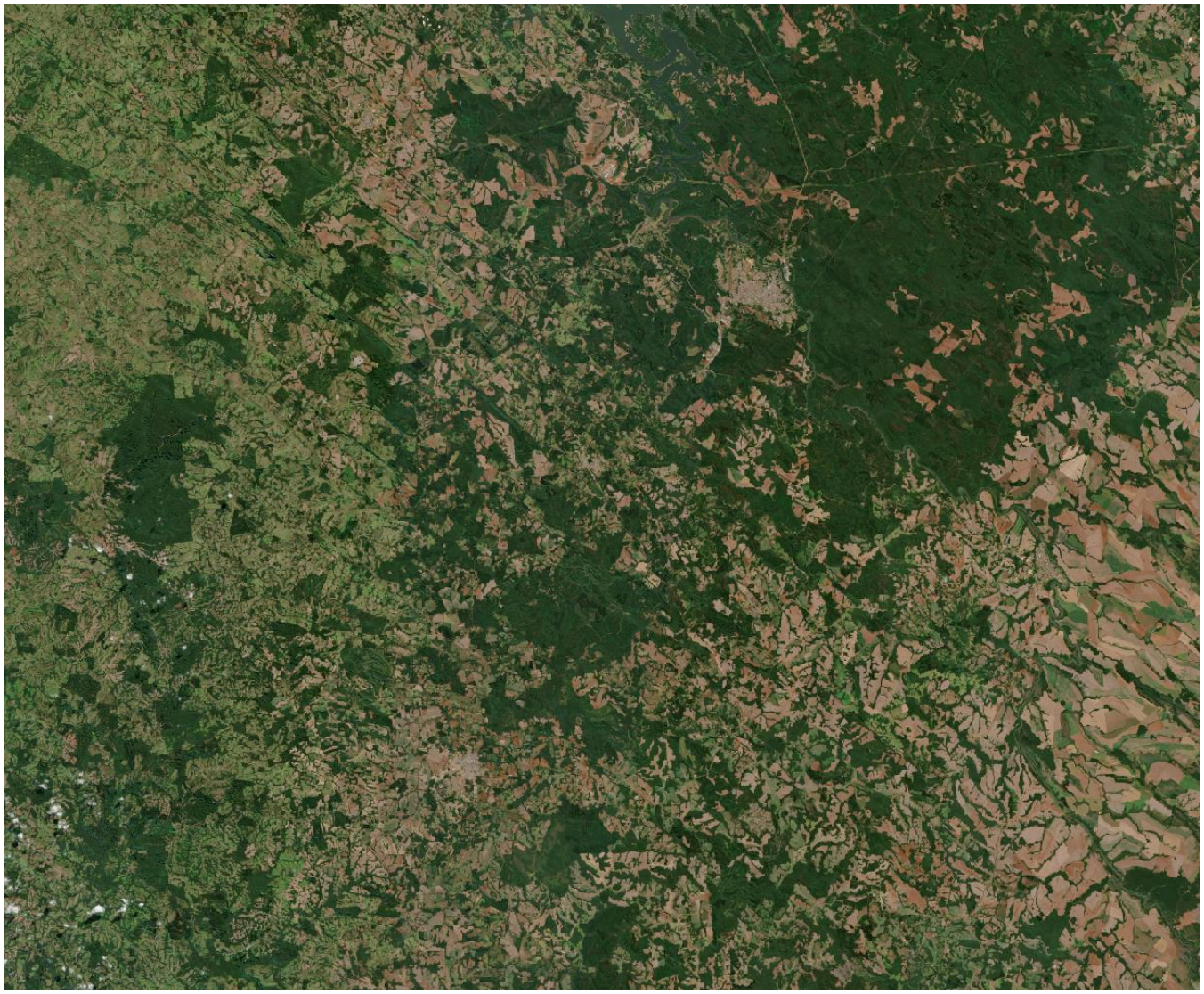
Class	Precision	Recall	F1-Score	Support
AnnualCrop	0.97	0.99	0.98	300
Forest	0.99	0.99	0.99	297
HerbVeg	0.98	0.96	0.97	302
Highway	0.96	0.99	0.97	269
Industrial	0.96	0.98	0.97	257
Pasture	0.97	0.98	0.98	187
PermCrop	0.96	0.97	0.96	254
Residential	1.00	0.97	0.98	314
River	0.99	0.95	0.97	245
SeaLake	0.99	0.99	0.99	275
Weighted avg	**0.98**	**0.98**	**0.98**	**2700**

STUDY AREA 01 — PARANÁ, BRAZIL

Atlantic Forest frontier *25.4°S, 51.9°W · 2018 - 2024*

Paraná's Atlantic Forest is one of the world's most threatened biomes, with less than 12% of original cover remaining. Agricultural expansion — particularly soy, maize, and cattle — has driven continued forest loss even after legal protections were strengthened. This scene covers a transition zone between remaining forest and active agricultural land.

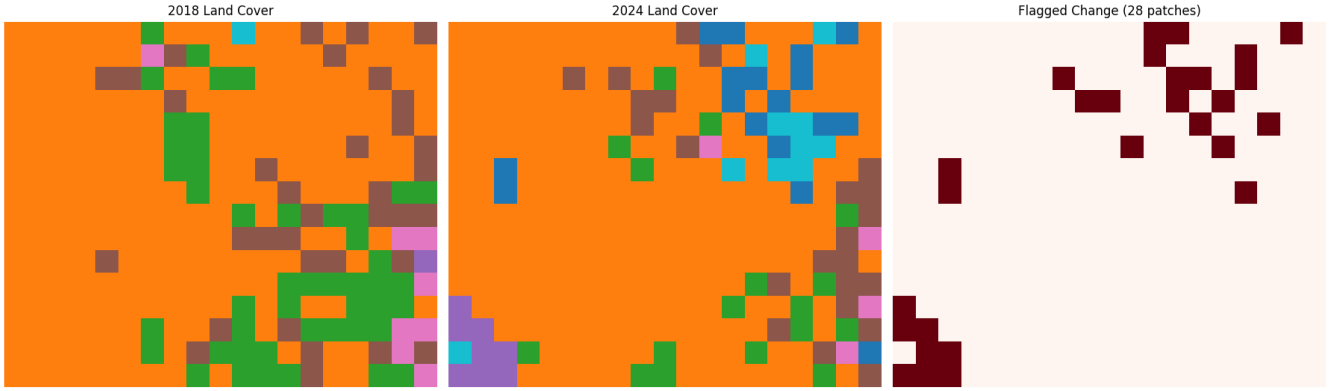
2018, Paraná, Brazil



2024, Paraná, Brazil



Flagged Changes:



KEY FINDINGS

- 28 patches flagged as Forest in 2018 and reclassified as AnnualCrop or Pasture in 2024.
- Flagged patches form coherent spatial clusters, consistent with agricultural clearing fronts rather than classification noise.
- Dominant background class in both years reflects the heavily agricultural character of the broader landscape.

STUDY AREA 02 — MATO GROSSO, BRAZIL

Amazon deforestation frontier *12.6°S, 55.7°W · 2018 - 2026*

Mato Grosso has historically recorded the highest absolute deforestation rates in the Brazilian Amazon, driven by soy agriculture and cattle ranching expanding into remaining Cerrado and forest. This scene covers a longer time window (2018–2026) to capture slower-moving agricultural frontier dynamics.

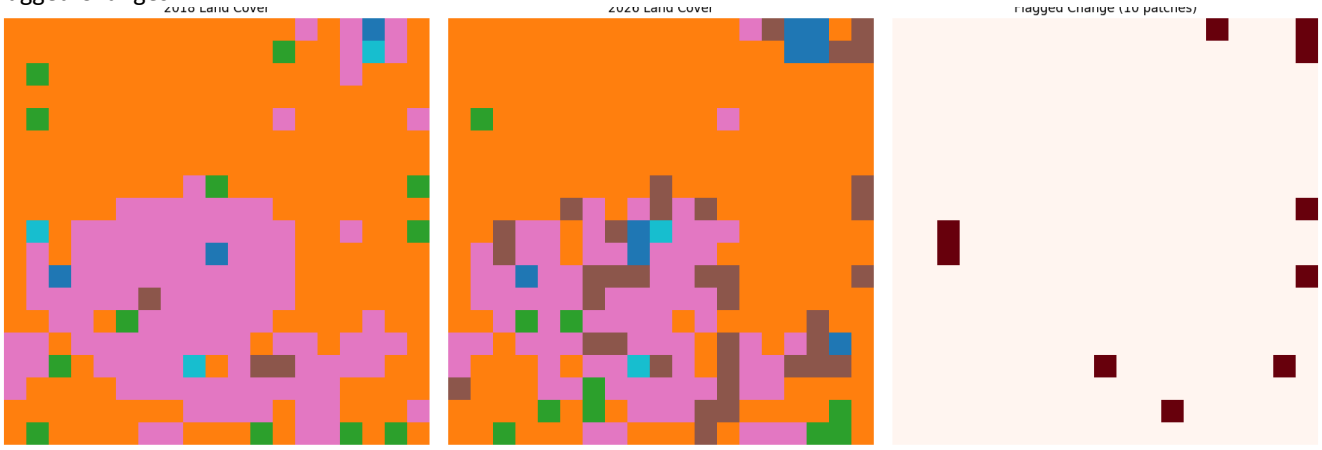
2018, Mato Grosso, Brazil



2026, Mato Grosso, Brazil



Flagged Changes:



KEY FINDINGS

- 10 patches flagged over an 8-year window meaning a lower count than Paraná, consistent with the scene covering a more interior, less immediately active clearing front.
- The dominant PermCrop class in the 2018 map reflects existing large-scale soy cultivation, with Forest patches present at the margins.
- Changes appear at the perimeter of existing agricultural areas, consistent with outward agricultural expansion.

STUDY AREA 03 — KARACHI, PAKISTAN

Peri-urban expansion *24.9°N, 67.1°E · 2018 - 2024*

Karachi is one of the world's most rapidly urbanizing megacities, with a population exceeding 20 million and significant peri-urban land conversion at its fringes. Unlike the Brazil sites, the expected change signal here is vegetation and open land converting to residential or industrial use rather than deforestation.

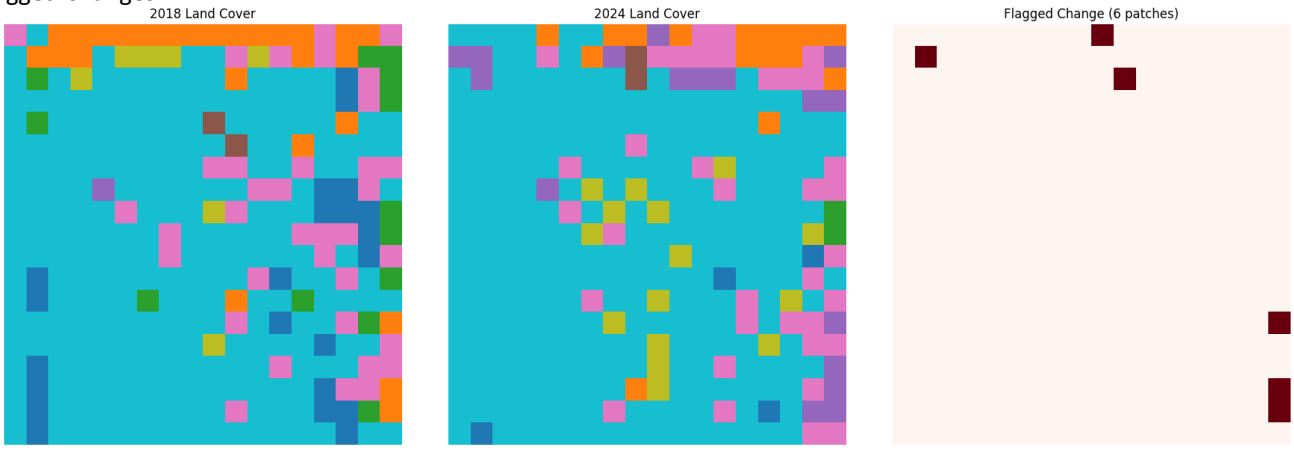
2018, Karachi, Pakistan



2024, Karachi, Pakistan



Flagged Changes:



KEY FINDINGS

- 6 flagged patches at the peri-urban fringe, consistent with outward residential and industrial encroachment.
- The dominant SeaLake/River class in both maps reflects the Arabian Sea and Karachi's coastal/creek geography which is correctly stable between dates.
- Fewer flagged patches than the Brazil sites, which may reflect the smaller footprint of per-patch urban conversion relative to agricultural clearing, or that the scene does not capture the highest-growth fringe areas.

LIMITATIONS & CRITICAL ANALYSIS

Several factors constrain the reliability of these results.

Patch Resolution

Each 64×64 patch at 10m/pixel covers ~640m × 640m. Sub-patch transitions like a narrow road cut, a small clearing etc. are invisible. The model sees one class per patch, not within-patch heterogeneity.

Seasonal Bias

No seasonal correction was applied. If the 2018 and 2024 images were acquired in different seasons, phenological differences (bare vs. vegetated fields) can generate false-positive change flags that are not real land-cover transitions.

Domain Shift

EuroSAT was built from European Sentinel-2 scenes. Tropical forest (Brazil) and South Asian semi-arid vegetation (Pakistan) have different spectral signatures. The model's 97.67% test accuracy may not transfer directly to these regions.

True Color Input

Imagery was exported as a visual True Color render, not raw reflectance bands. This bakes in a contrast stretch applied by the Copernicus Browser, potentially reducing radiometric comparability across dates if the stretch parameters differ.

Single Image per Date

Using the least-cloudy single acquisition rather than a cloud-masked median composite means any remaining cloud shadows or haze can be misclassified. A composited time window would reduce this risk.

No Ground Truth

The flagged transitions have not been validated against independent ground truth (GPS surveys, government deforestation records, or PRODES data for Brazil). Precision and recall of the change detection step are unknown.

REFERENCES

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